

"The Random Variable is Roughly Its Expectation" — The Kahn-Kalai Conjectures

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1 Basic setup

Let X be a finite set and let $p \in (0, 1)$. We write

$$|X| < \infty.$$

For each $p \in (0, 1)$, define the random subset $X_p \subseteq X$ by including each $\omega \in X$ independently with probability p . Equivalently,

$$X_p = \{\omega \in X : \omega \text{ is selected}\},$$

where the selections are independent and

$$\mathbb{P}(\omega \in X_p) = p, \quad \omega \in X.$$

The law of X_p is therefore the product measure on 2^X given by

$$\mathbb{P}(X_p = S) = p^{|S|}(1-p)^{|X|-|S|}, \quad S \subseteq X.$$

2 Increasing families

A family $\mathcal{F} \subseteq 2^X$ is called *increasing* if

$$S \in \mathcal{F}, \quad S \subseteq S' \implies S' \in \mathcal{F}.$$

In words, once a set belongs to $\mathcal{F}$, every superset also belongs to $\mathcal{F}$.

For an increasing family $\mathcal{F}$, the quantity

$$\mathbb{P}(X_p \in \mathcal{F})$$

is nondecreasing as a function of p . In particular, if $0 < p_1 \leq p_2 < 1$, then

$$\mathbb{P}(X_{p_1} \in \mathcal{F}) \leq \mathbb{P}(X_{p_2} \in \mathcal{F}).$$

Remark 1. *The monotonicity follows by coupling the product measures: take independent $U_\omega \sim \text{Unif}(0, 1)$ and define*

$$X_p = \{\omega \in X : U_\omega \leq p\}.$$

Then $p_1 \leq p_2$ implies $X_{p_1} \subseteq X_{p_2}$ almost surely. Since $\mathcal{F}$ is increasing,

$$\mathbf{1}_{\{X_{p_1} \in \mathcal{F}\}} \leq \mathbf{1}_{\{X_{p_2} \in \mathcal{F}\}} \quad \text{a.s.}$$

Taking expectations gives the claimed inequality.

3 Thresholds for increasing properties

By monotonicity of an increasing family, $\mu_{\mathcal{F}}(p)$ is nondecreasing in p .

3.1 Threshold scale

The board introduces a sequence of increasing families $(\mathcal{F}_n)$ and a sequence of parameters (p_n) . The intended threshold notion is expressed by comparing a parameter q_n with p_n :

$$q_n = o(p_n) \implies \mathbb{P}(X_{n,q_n} \in \mathcal{F}_n) \longrightarrow 0,$$

whereas

$$q_n = \omega(p_n) \implies \mathbb{P}(X_{n,q_n} \in \mathcal{F}_n) \longrightarrow 1.$$

Here $X_{n,q}$ denotes the q -random subset of the n th ground set.

A common normalization is the median threshold

$$p_n^* := \inf \left\{ p \in [0, 1] : \mathbb{P}(X_{n,p} \in \mathcal{F}_n) \geq \frac{1}{2} \right\}.$$

For increasing properties, the monotonicity of $\mu_{\mathcal{F}_n}$ makes this a natural location at which to measure the transition from probability close to 0 to probability close to 1.

3.2 Example: consecutive elements

Consider the increasing property

$$\mathcal{F}_n = \{A \subseteq [n] : A \text{ contains two consecutive integers}\}.$$

Introduce the indicator variables

$$I_i := \mathbf{1}_{\{i, i+1 \in X_p\}}, \quad i = 1, \dots, n-1,$$

and the count

$$Y := \sum_{i=1}^{n-1} I_i.$$

Then

$$\{X_p \in \mathcal{F}_n\} = \{Y \geq 1\}.$$

Since $\mathbb{E}I_i = p^2$, linearity gives

$$\mathbb{E}Y = \sum_{i=1}^{n-1} \mathbb{E}I_i = (n-1)p^2.$$

Consequently, Markov's inequality yields

$$\mathbb{P}(X_p \in \mathcal{F}_n) = \mathbb{P}(Y \geq 1) \leq \mathbb{E}Y = (n-1)p^2.$$

In particular, if $np^2 \rightarrow 0$, then

$$\mathbb{P}(X_p \in \mathcal{F}_n) \rightarrow 0.$$

For the second moment, write

$$Y^2 = \sum_{i=1}^{n-1} I_i + 2 \sum_{1 \leq i < j \leq n-1} I_i I_j.$$

If $|i - j| \geq 2$, the pairs $\{i, i + 1\}$ and $\{j, j + 1\}$ are disjoint, so

$$\mathbb{E}(I_i I_j) = p^4.$$

If $j = i + 1$, the two events overlap in one element and hence

$$\mathbb{E}(I_i I_{i+1}) = p^3.$$

Therefore

$$\mathbb{E}Y^2 = (n - 1)p^2 + 2(n - 2)p^3 + ((n - 1)(n - 2) - 2(n - 2))p^4,$$

where the last term counts the non-overlapping pairs. Equivalently, one may keep the decomposition in terms of overlapping and disjoint pairs before simplifying.

3.3 Existence of a Threshold

Let $\mathcal{F}_n \subseteq 2^{X_n}$ be increasing, and define

$$f_n(p) := \mathbb{P}(X_{n,p} \in \mathcal{F}_n), \quad p_n := \inf \left\{ p : f_n(p) \geq \frac{1}{2} \right\}.$$

Suppose that

$$\frac{p_n}{q_n} \longrightarrow \infty.$$

Choose integers K_n such that

$$K_n < \frac{p_n}{q_n}, \quad K_n \longrightarrow \infty.$$

Let $X_{n,q_n}^{(1)}, \dots, X_{n,q_n}^{(K_n)}$ be independent copies of X_{n,q_n} , and put

$$U_n := \bigcup_{i=1}^{K_n} X_{n,q_n}^{(i)}.$$

Then

$$U_n \stackrel{d}{=} X_{n, 1 - (1 - q_n)^{K_n}}.$$

Since

$$1 - (1 - q_n)^{K_n} \leq K_n q_n < p_n,$$

we may couple $U_n \subseteq X_{n,p_n}$ almost surely. Hence, by monotonicity,

$$\mathbb{P}(U_n \in \mathcal{F}_n) \leq f_n(p_n) \leq \frac{1}{2}.$$

On the other hand, writing $f_n(q_n) = \mathbb{P}(X_{n,q_n} \in \mathcal{F}_n)$, independence gives

$$\begin{aligned} \frac{1}{2} &\geq \mathbb{P}(U_n \in \mathcal{F}_n) \\ &\geq \mathbb{P} \left(\bigcup_{i=1}^{K_n} \{X_{n,q_n}^{(i)} \in \mathcal{F}_n\} \right) \\ &= 1 - (1 - f_n(q_n))^{K_n}. \end{aligned}$$

Therefore,

$$f_n(q_n) \leq 1 - 2^{-1/K_n} \leq \frac{\log 2}{K_n} \longrightarrow 0.$$

Thus

$$\boxed{\frac{p_n}{q_n} \rightarrow \infty \implies f_n(q_n) \rightarrow 0.}$$

Similarly, one can show the other side.

4 The Kahn–Kalai Conjectures

Let $\mathcal{F} \subseteq 2^X$ be an increasing family, where X is finite. For $\mathcal{G} \subseteq 2^X$, define

$$\langle \mathcal{G} \rangle := \{A \subseteq X : \exists B \in \mathcal{G} \text{ such that } B \subseteq A\}.$$

We say that $\mathcal{G}$ is a *cover* of $\mathcal{F}$ if

$$\mathcal{F} \subseteq \langle \mathcal{G} \rangle.$$

For $p \in [0, 1]$, let X_p be a random subset of X , where each element is present independently with probability p . The expected number of members of $\mathcal{G}$ present in X_p is

$$\mathbb{E}[|\{B \in \mathcal{G} : B \subseteq X_p\}|] = \sum_{B \in \mathcal{G}} p^{|B|}.$$

Define the cost of $\mathcal{F}$ at p by

$$\text{cost}_p(\mathcal{F}) := \inf_{\mathcal{G} : \mathcal{F} \subseteq \langle \mathcal{G} \rangle} \sum_{B \in \mathcal{G}} p^{|B|}.$$

The *expectation threshold* is

$$q(\mathcal{F}) := \sup \left\{ p \in [0, 1] : \text{cost}_p(\mathcal{F}) \leq \frac{1}{2} \right\}.$$

Let

$$p_c(\mathcal{F}) := \inf \left\{ p \in [0, 1] : \mathbb{P}(X_p \in \mathcal{F}) \geq \frac{1}{2} \right\},$$

and let

$$\ell(\mathcal{F}) := \max \left\{ 2, \max_{A \in \mathcal{F}_{\min}} |A| \right\},$$

where $\mathcal{F}_{\min}$ is the family of minimal members of $\mathcal{F}$.

4.1 KKC1: The Kahn–Kalai Conjecture (Park–Pham Theorem).

There exists a universal constant $K > 0$ such that

$$\boxed{p_c(\mathcal{F}) \leq K q(\mathcal{F}) \log \ell(\mathcal{F}).}$$

4.2 KKC2: The Graph-Containment Conjecture.

Let H be a fixed graph and let N be the number of vertices of the ambient graph K_N . Let $G_{N,p}$ denote the Erdős–Rényi random graph on vertex set $[N]$, in which every edge is present independently with probability p .

Define the expectation threshold $p_E(H)$ to be the smallest p such that

$$\mathbb{E}[\#\{\text{copies of } H' \text{ in } G_{N,p}\}] \geq 1$$

for every subgraph $H' \subseteq H$.

Then there exists a universal constant $K > 0$ such that

$$p_c(H) \leq K p_E(H) \log N,$$

where

$$p_c(H) = \inf \left\{ p : \mathbb{P}(G_{N,p} \text{ contains a copy of } H) \geq \frac{1}{2} \right\}.$$

Remark 2. *The second Kahn–Kalai conjecture is stronger than the first because it concerns the more general setting of graph containment. Indeed, KKC1 applies to arbitrary increasing families of subsets, whereas KKC2 applies to the increasing property of containing a given graph H and relates its threshold to the expectation threshold of every subgraph of H . Thus, KKC2 requires a bound in terms of a more specific and potentially smaller parameter, making it a stronger assertion than KKC1.*

5 A short proof of the Kahn–Kalai conjecture

For any family $\mathcal{F} \subseteq 2^X$, define its upward closure by

$$\langle \mathcal{F} \rangle = \{T \subseteq X : \text{there exists } S \in \mathcal{F} \text{ such that } S \subseteq T\}.$$

We say that $\mathcal{G}$ covers $\mathcal{F}$ if

$$\mathcal{F} \subseteq \langle \mathcal{G} \rangle.$$

For a family $\mathcal{F}$, define its covering cost by

$$f_p(\mathcal{F}) = \inf_{\mathcal{G}: \mathcal{F} \subseteq \langle \mathcal{G} \rangle} E_p(\mathcal{G}).$$

The cost function is subadditive: if

$$\mathcal{F} = \mathcal{F}_1 \cup \mathcal{F}_2,$$

then

$$f_p(\mathcal{F}) \leq f_p(\mathcal{F}_1) + f_p(\mathcal{F}_2).$$

Moreover,

$$f_p(\emptyset) = 0, \quad f_p(\mathcal{F}) = 1 \quad \text{if } \emptyset \in \mathcal{F}.$$

For $0 \leq m \leq N$, let

$$\mathcal{L}_m = \binom{X}{m} = \{S \subseteq X : |S| = m\}$$

be the m -th level of the Boolean lattice. Thus,

$$|\mathcal{L}_m| = \binom{N}{m}.$$

If $\mathcal{G} \subseteq 2^X$ is increasing, define

$$c_m(\mathcal{G}) = \frac{|\mathcal{G} \cap \mathcal{L}_m|}{|\mathcal{L}_m|}.$$

By a standard double-counting argument,

$$c_m(\mathcal{G})$$

is increasing in m .

5.1 The covering theorem

For a positive integer ℓ , say that a family $\mathcal{F}$ is ℓ -bounded if

$$|S| \leq \ell \quad \text{for every } S \in \mathcal{F}.$$

Let

$$m_\ell = \lfloor LpN \log(\ell + 1) \rfloor,$$

where L is a sufficiently large absolute constant.

Theorem 1 (Covering theorem). *There exists an absolute constant $L > 0$ such that the following holds.*

Let $\mathcal{F} \subseteq 2^X$ be an ℓ -bounded family, where $|X| = N$, and suppose that

$$f_p(\mathcal{F}) \geq \frac{1}{2} - \frac{1}{2^{\ell+2}}.$$

Then

$$c_{m_\ell}(\langle \mathcal{F} \rangle) \geq \frac{2}{3} + \frac{1}{2^{\ell+2}}.$$

Equivalently,

$$|\langle \mathcal{F} \rangle \cap \mathcal{L}_{m_\ell}| \geq \left(\frac{2}{3} + \frac{1}{2^{\ell+2}} \right) |\mathcal{L}_{m_\ell}|.$$

Proof. We prove the theorem by induction on ℓ and N .

Base case: $\ell = 0$. Since $\mathcal{F}$ is 0-bounded, every member of $\mathcal{F}$ is the empty set. The assumption

$$f_p(\mathcal{F}) \geq \frac{1}{2} - \frac{1}{4} = \frac{1}{4} > 0$$

therefore implies that

$$\emptyset \in \mathcal{F}.$$

Consequently,

$$\langle \mathcal{F} \rangle = 2^X,$$

and hence

$$c_{m_0}(\langle \mathcal{F} \rangle) = 1.$$

The conclusion follows.

Induction step. Assume that the result holds for every $\ell' < \ell$ and every ground-set size $N' \leq N$.

For a set $W \subseteq X$, define

$$\mathcal{F}_W = \{S \setminus W : S \in \mathcal{F}\}.$$

Let $\mathcal{F}'_W$ be the family of inclusion-minimal members of $\mathcal{F}_W$.

Define

$$\mathcal{G}_W = \{T \in \mathcal{F}'_W : |T| > 0.9\ell\},$$

and

$$\tilde{\mathcal{F}}_W = \mathcal{F}'_W \setminus \mathcal{G}_W.$$

Thus, every member of $\tilde{\mathcal{F}}_W$ has size at most

$$\ell_1 := \lfloor 0.9\ell \rfloor < \ell.$$

We first prove the following double-counting lemma.

Lemma 1 (Double-counting lemma). *Suppose that $L > 1000$ and set*

$$w = \lfloor 0.1LpN \rfloor.$$

Then

$$\sum_{\substack{W \subseteq X \\ |W|=w}} f_p(\mathcal{G}_W) \leq \binom{N}{w} \frac{1}{8 \cdot 16^\ell}.$$

Proof. By the definition of the covering cost,

$$f_p(\mathcal{G}_W) \leq E_p(\mathcal{G}_W) = \sum_{S' \in \mathcal{G}_W} p^{|S'|}.$$

Therefore,

$$\begin{aligned} \sum_{|W|=w} f_p(\mathcal{G}_W) &\leq \sum_{|W|=w} \sum_{S' \in \mathcal{G}_W} p^{|S'|} \\ &= \sum_k \sum_{\substack{|W|=w \\ S' \in \mathcal{G}_W \\ |S'|=k}} p^k. \end{aligned}$$

We now bound the number of pairs (W, S') appearing in the last sum, where

$$|W| = w, \quad S' \in \mathcal{G}_W, \quad |S'| = k.$$

Fix such a pair and put

$$W' = W \cup S'.$$

Since $S' \subseteq X \setminus W$, we have

$$|W'| = w + k.$$

There are

$$\binom{N}{w+k}$$

possibilities for W' .

Fix W' . Since $S' \in \mathcal{F}'_W$, there exists some $S \in \mathcal{F}$ such that

$$S \setminus W = S'.$$

In particular,

$$S' \subseteq W'.$$

We claim that

$$S' \subseteq S.$$

Indeed, if $S' \not\subseteq S$, then

$$S' \cap S = S \setminus W$$

would be a proper subset of S' . But $S' \cap S$ belongs to $\mathcal{F}_W$, contradicting the minimality of S' in $\mathcal{F}'_W$.

Since $S \in \mathcal{F}$ and $\mathcal{F}$ is ℓ -bounded,

$$|S| \leq \ell.$$

Hence, after fixing S , there are at most

$$\binom{\ell}{k}$$

possible choices for S' .

Thus,

$$\#\{(W, S') : |W| = w, S' \in \mathcal{G}_W, |S'| = k\} \leq \binom{N}{w+k} \binom{\ell}{k}.$$

Because $S' \in \mathcal{G}_W$, we have

$$0.9\ell < k \leq \ell.$$

Consequently,

$$\begin{aligned} \sum_{|W|=w} f_p(\mathcal{G}_W) &\leq \sum_{0.9\ell < k \leq \ell} \binom{N}{w+k} \binom{\ell}{k} p^k \\ &\leq \binom{N}{w} \sum_{0.9\ell < k \leq \ell} (0.1L)^{-k} \binom{\ell}{k}. \end{aligned}$$

Here we used

$$w = \lfloor 0.1LpN \rfloor$$

and the standard comparison

$$\binom{N}{w+k} p^k \leq \binom{N}{w} (0.1L)^{-k}.$$

Since $k > 0.9\ell$,

$$(0.1L)^{-k} \leq (0.1L)^{-0.9\ell}.$$

Therefore,

$$\begin{aligned} \sum_{|W|=w} f_p(\mathcal{G}_W) &\leq \binom{N}{w} (0.1L)^{-0.9\ell} \sum_{k=0}^{\ell} \binom{\ell}{k} \\ &= \binom{N}{w} (0.1L)^{-0.9\ell} 2^\ell. \end{aligned}$$

For $L \geq 1000$,

$$(0.1L)^{-0.9\ell} 2^\ell \leq \frac{1}{8 \cdot 16^\ell}.$$

Hence,

$$\sum_{|W|=w} f_p(\mathcal{G}_W) \leq \binom{N}{w} \frac{1}{8 \cdot 16^\ell}.$$

This proves the lemma. □

We now return to the proof of the covering theorem.

Call a set $W \subseteq X$ with $|W| = w$ *good* if

$$f_p(\mathcal{G}_W) \leq \frac{1}{2^{\ell+2}},$$

and *bad* otherwise.

By the double-counting lemma and Markov's inequality, the fraction of bad sets W is at most

$$\frac{\frac{1}{8 \cdot 16^\ell}}{\frac{1}{2^{\ell+2}}} = \frac{1}{2 \cdot 8^\ell}.$$

Thus, at least the fraction

$$1 - \frac{1}{2 \cdot 8^\ell}$$

of all W are good.

Fix a good set W . Since $\mathcal{F}'_W$ is a family of minimal members of $\mathcal{F}_W$, we have

$$\langle \mathcal{F}'_W \rangle = \langle \mathcal{F}_W \rangle.$$

Moreover,

$$\mathcal{F}'_W = \tilde{\mathcal{F}}_W \cup \mathcal{G}_W.$$

By subadditivity,

$$f_p(\mathcal{F}'_W) \leq f_p(\tilde{\mathcal{F}}_W) + f_p(\mathcal{G}_W).$$

Consequently,

$$\begin{aligned} f_p(\tilde{\mathcal{F}}_W) &\geq f_p(\mathcal{F}'_W) - f_p(\mathcal{G}_W) \\ &\geq f_p(\mathcal{F}) - \frac{1}{2^{\ell+2}} \\ &\geq \frac{1}{2} - \frac{1}{2^{\ell+1}}. \end{aligned}$$

Since

$$\ell_1 = \lfloor 0.9\ell \rfloor < \ell,$$

we have

$$\frac{1}{2} - \frac{1}{2^{\ell+1}} \geq \frac{1}{2} - \frac{1}{2^{\ell_1+2}}.$$

Therefore, the induction hypothesis applies to $\tilde{\mathcal{F}}_W$, which is ℓ_1 -bounded on the ground set $X \setminus W$.

It follows that

$$\langle \tilde{\mathcal{F}}_W \rangle$$

contains at least the fraction

$$\frac{2}{3} + \frac{1}{2^{\ell_1+2}}$$

of the

$$m_{\ell_1} = \lfloor Lp(N-w) \log(\ell_1 + 1) \rfloor$$

level of the Boolean lattice on $X \setminus W$.

Taking the union of each such set with W , we see that $\langle \mathcal{F} \rangle$ contains at least

$$\left(\frac{2}{3} + \frac{1}{2^{\ell_1+2}} \right) \left(1 - \frac{1}{2 \cdot 8^\ell} \right)$$

fraction of the

$$(m_{\ell_1} + w)$$

level of the Boolean lattice on X .

Now,

$$\begin{aligned} m_{\ell_1} + w &= \lfloor Lp(N-w) \log(\ell_1 + 1) \rfloor + \lfloor 0.1LpN \rfloor \\ &\leq \lfloor LpN \log(\ell + 1) \rfloor \\ &= m_\ell, \end{aligned}$$

provided L is sufficiently large.

Since $\langle \mathcal{F} \rangle$ is increasing, its density on levels is increasing. Hence,

$$\begin{aligned} c_{m_\ell}(\langle \mathcal{F} \rangle) &\geq \left(\frac{2}{3} + \frac{1}{2^{\ell_1+2}} \right) \left(1 - \frac{1}{2 \cdot 8^\ell} \right) \\ &\geq \frac{2}{3} + \frac{1}{2^{\ell+2}}. \end{aligned}$$

Finally, since $\mathcal{F}$ is increasing, we have

$$\begin{aligned} \mathbb{P}(X_p \in \mathcal{F}) &\geq \mathbb{P}(|X_p| \geq m_\ell) c_{m_\ell}(\mathcal{F}) \\ &\geq \frac{2}{3} \mathbb{P}(|X_p| \geq m_\ell) \\ &\geq \frac{2}{3} \cdot \frac{3}{4} \\ &= \frac{1}{2}. \end{aligned}$$

Here we have used the estimate

$$\mathbb{P}(|X_p| \geq m_\ell) \geq \frac{3}{4}.$$

Therefore,

$$\boxed{\mathbb{P}(X_p \in \mathcal{F}) \geq \frac{1}{2}.$$

This proves the theorem. □